

VOLT SENTINEL

Smart Last-Mile Power Line Safety & Breakage Detection

Problem Statement ID	SIH25063
Theme	Disaster Management
PS Category	Hardware
Team Name	GridNova
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This report summarizes and expands on the GridNova team's Smart India Hackathon 2026 submission for Problem Statement SIH25063 — Volt Sentinel, a low-cost sensor network for detecting power line breakage and faults along last-mile electricity board (EB) distribution feeders.

Report prepared from the team's SIH idea-submission slide deck.

1. Executive Summary

Last-mile power lines run for kilometres between substations and end consumers, and faults such as snapped conductors, insulation leakage, or tree contact are usually found only after a complaint or a manual line-walk covering the entire stretch. Volt Sentinel proposes a distributed sensing approach: a chain of low-cost ESP32-based nodes, each fitted with a current and voltage sensor, is installed along the feeder. By continuously comparing electrical readings between neighbouring nodes, the system narrows a fault down to the specific span between two nodes rather than the full length of the line, cutting diagnosis time and the distance line crews must patrol.

2. Problem Context

The submission is framed as a hardware solution under the Disaster Management theme, addressing the risks created by undetected power line faults: prolonged outages, fire hazards from arcing or leakage, and safety risks to the public and to maintenance crews sent out to search a long feeder for an unknown fault location. The reference scenario used in the deck is a 10 km EB line instrumented with 20 sensor nodes spaced along its length.

Core idea at a glance:

- 20 sensor nodes deployed along a 10 km EB feeder.
- Each node prototype pairs a current sensor and a voltage sensor with an ESP32.
- Nodes stream continuous telemetry over Wi-Fi to the EB office / server.
- The server compares readings between adjacent nodes to localize faults.

3. Proposed Solution

Volt Sentinel works by giving every segment of the feeder its own point of measurement, then reasoning about differences between neighbouring points instead of relying on a single point-of-failure sensor at the substation.

3.1 Deployment Layout

- 20 sensor nodes are distributed along the 10 km EB line at roughly even intervals.
- Each node continuously reports its current and voltage readings to the EB office.

3.2 Fault Localization Logic

The system's central logic is differential comparison between adjacent nodes:

- If nodes further down the line stop reporting while earlier nodes remain active, the break is inferred to lie in the span between the last responding node and the first silent one. In the deck's example, Nodes 1–15 stay active while Nodes 16–20 go offline, pointing to a fault between Node 15 and Node 16.
- If two adjacent nodes that are both online show a meaningful difference in voltage or current, that difference is treated as a sign of leakage or a partial fault such as tree contact at that specific span, even though the line has not fully broken.

3.3 Why This Is Faster Than Manual Inspection

A conventional response to an outage or reported fault involves a crew walking or driving the entire feeder to visually locate the problem. By narrowing the search to a single inter-node span, Volt Sentinel is designed to convert what is normally a kilometres-long search into a section-level lookup, giving section-level detection instead of full-line detection.

4. Technical Approach

4.1 Hardware

- Current sensor — measures line current at each node position.
- Voltage sensor — measures line voltage at each node position.
- ESP32 microcontroller — reads both sensors and handles Wi-Fi communication.
- Power supply — powers the node electronics in the field.
- Protected enclosure — houses the electronics against outdoor conditions.

4.2 Communication

Each ESP32 node communicates over Wi-Fi, sending readings to a central server or the EB office. The deck notes ESP32 Wi-Fi as supporting high-speed, real-time telemetry using WebSocket and HTTP REST style communication.

4.3 Data Flow

The processing pipeline follows four stages: **Sampling** → **Transmission** → **Node Comparison** → **Fault Alert**.

- **Sampling:** each ESP32 node measures voltage and current at its location.
- **Transmission:** readings are sent to the central server as continuous telemetry.
- **Node comparison:** the server checks node status (online/offline) and signal continuity across the chain, and runs differential analysis between adjacent nodes to identify the affected span.
- **Fault alert:** a dashboard raises an alert carrying the fault's location, fault type, and time.

This pipeline is what lets the system move from raw per-node electrical readings to an actionable, location-specific alert for the EB operations team.

5. Feasibility and Viability

The design deliberately favours low-cost, readily available components (basic current/voltage sensors and the ESP32 module) over specialized or expensive monitoring hardware, and the modular per-node structure means additional nodes can be added to cover longer feeders without redesigning the system. Wi-Fi-based node-to-server communication is highlighted as giving low-latency data transmission, which matters for near-real-time fault detection.

5.1 Identified Risks and Controls

Risk	Control
Weather exposure	Sealed outdoor enclosure
Lightning	Surge protection and proper earthing
Weak Wi-Fi signal	Auto-reconnect routines
Sensor error / drift	Calibration and plausibility checks
False alarms	Persistence timer plus adjacent-node validation

Pilot path: the rollout plan moves from a bench test, to a small 3-node trial, to full 20-node field deployment — allowing the team to validate the sensing and alerting logic at increasing scale before committing to full installation.

6. Impact and Benefits

Target users: EB operators, maintenance teams, industrial customers, and reliability planners.

- Reduces the fault search area from the full 10 km line to a single affected span.
- Shortens outage diagnosis and restoration time.
- Helps identify leakage points and vegetation (tree) contact points on the line.
- Reduces the distance and exposure time for maintenance crews patrolling the feeder.
- Supports prevention of electrical fires and other unsafe conditions caused by undetected faults.
- Improves reliability reporting and the quality of updates given to customers.

Proposed metrics: node availability, alert time, false-alarm rate, patrol distance saved, and restoration time.

7. Research and References

The submission grounds its design in existing safety and reliability standards rather than an ad hoc approach:

- **CEA Safety Regulations, 2023** — covers electrical safety, isolation, earthing, and overhead-line work practices, informing the enclosure, earthing, and surge-protection choices.
- **IEEE C37.114-2014** — provides guidance on AC transmission and distribution fault-location methods, relevant to the node-comparison fault-localization logic.
- **IEEE 1366-2022** — defines distribution reliability indices, which inform the metrics proposed for evaluating the system (outage/restoration time, reliability reporting).
- **ESP32 Wi-Fi protocol documentation** — supports the choice of WebSocket / HTTP REST telemetry for real-time node-to-server communication.

Reference list (as cited in the submission):

- [1] CEA Safety and Electric Supply Regulations
- [2] ESP32 Wi-Fi & TCP/IP Networking Documentation
- [3] IEEE C37.114-2014
- [4] IEEE 1366-2022

8. Conclusion

Volt Sentinel's core contribution is a simple but effective idea: rather than monitoring a feeder as a single unit, break it into instrumented spans and let differences between neighbouring nodes reveal both hard breaks and softer faults like leakage or tree contact. The approach uses inexpensive, off-the-shelf hardware (ESP32 plus current/voltage sensors), a staged pilot plan to de-risk deployment, and named safety and reliability standards to ground its design choices. If the differential-comparison logic and false-alarm controls hold up under field conditions, the main payoff is a large reduction in the time and distance needed to find and respond to line faults compared with manual inspection.